Spark Internals

Spark SQL and the Underlying Engine

- Unifies Spark components and permits abstraction to DataFrames/Datasets in Java, Scala, Python, and R, which simplifies working with structured data sets.
- Connects to the Apache Hive metastore and tables.
- Reads and writes structured data with a specific schema from structured file formats (JSON, CSV, Text, Avro, Parquet, ORC, etc.) and converts data into temporary tables.
- Offers an interactive Spark SQL shell for quick data exploration.
- Provides a bridge to (and from) external tools via standard database JDBC/ODBC connectors.
- Generates optimized query plans and compact code for the JVM, for final execution.

Spark SQL and its stack

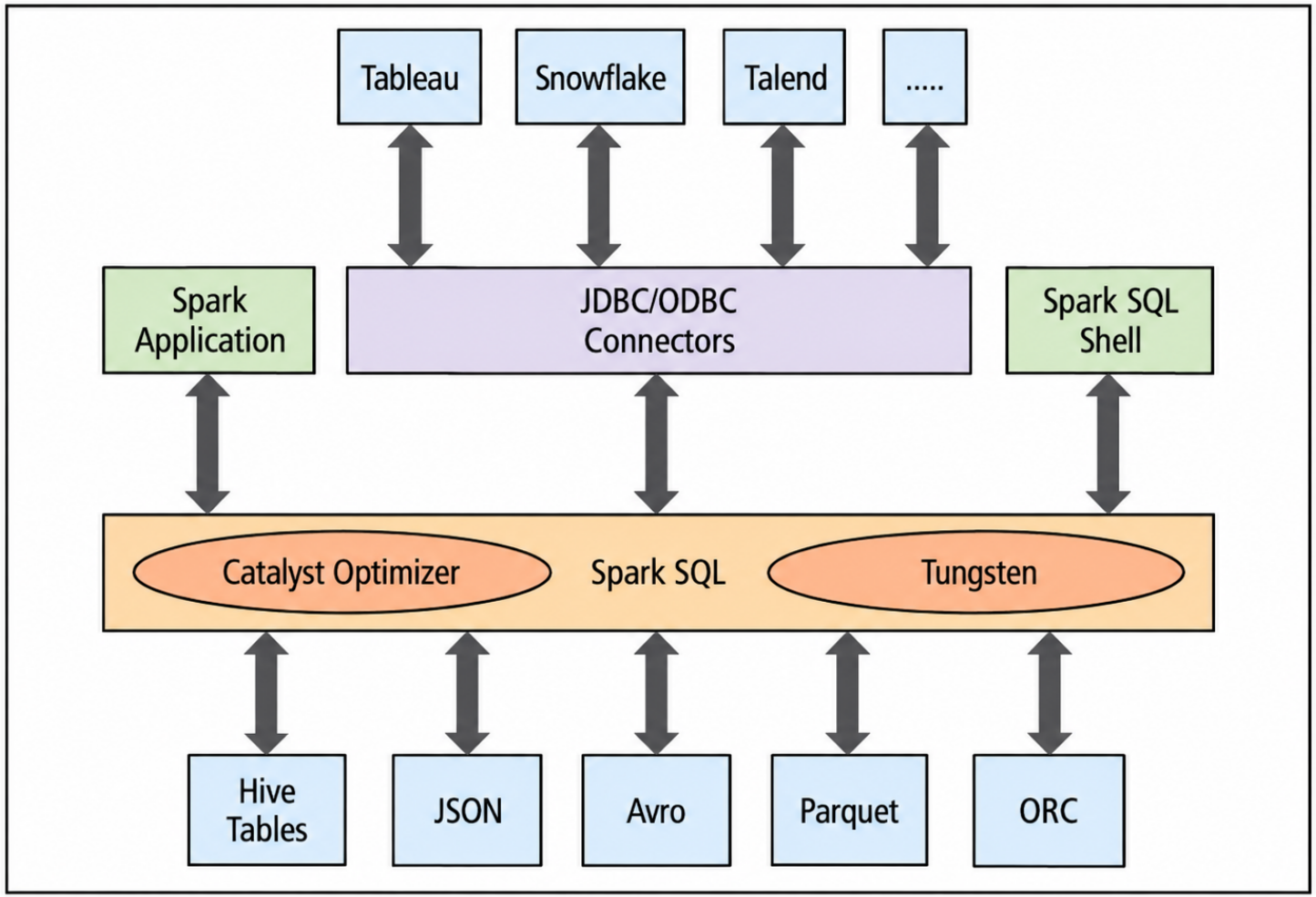


Figure 3-3. Spark SQL and its stack

The Catalyst Optimizer

The Catalyst optimizer takes a computational query and converts it into an execution plan. It goes through four transformational phases.

- Analysis
- Logical optimization
- Physical planning
- Code generation

Journey of a Spark Computation

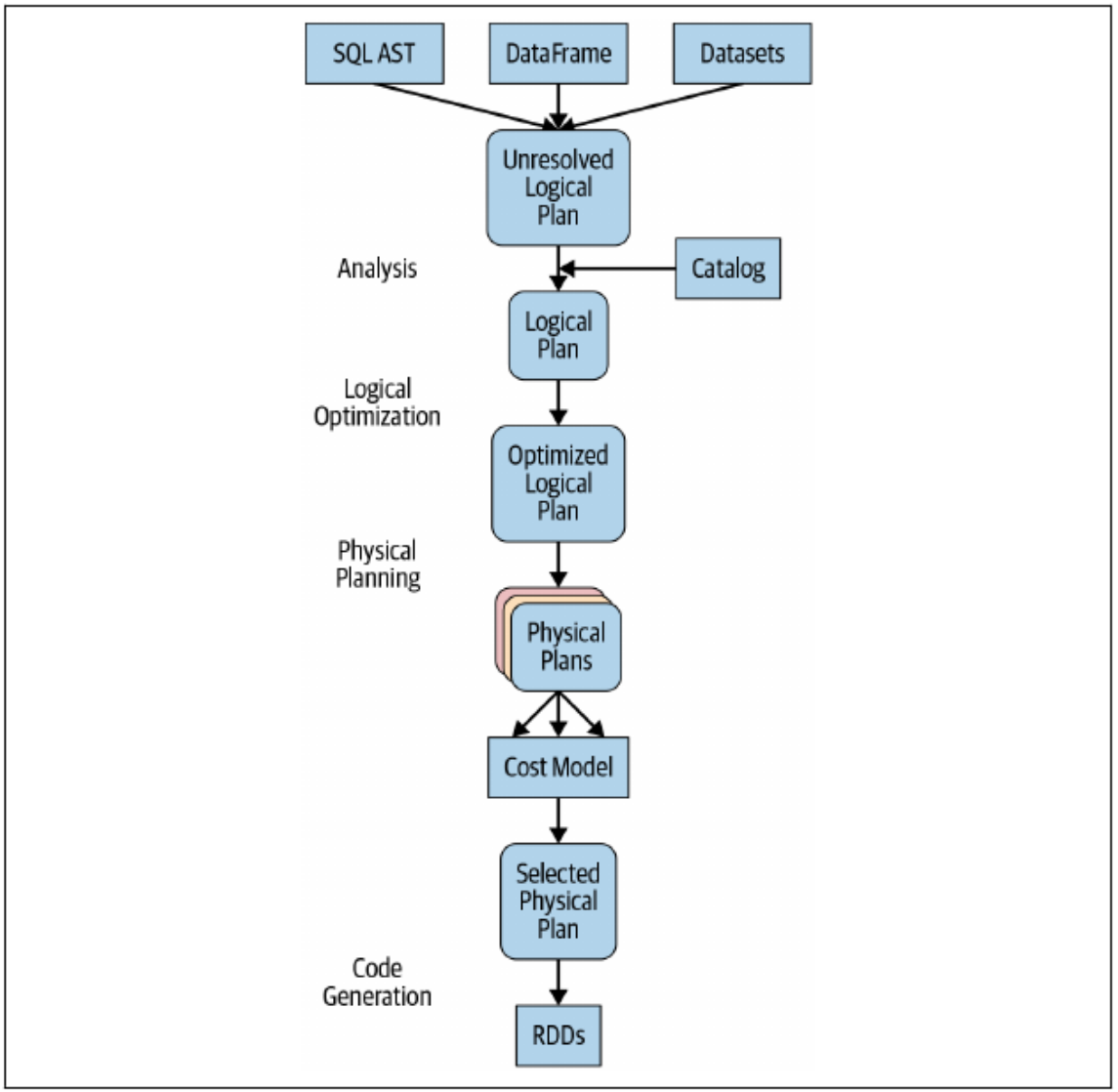


Figure 3-4. A Spark computation's four-phase journey

Notes: SQL AST denotes SQL Abstract Syntax Tree